

Parsa Hosseini

Contact: phoseini@umd.edu Links: Website LinkedIn Google Scholar

EDUCATION

- **University of Maryland** College Park, Maryland
PhD in Computer Science, Advisor: Soheil Feizi Aug 2024 - Present
- **University of Maryland** College Park, Maryland
MSc in Computer Science, GPA: 4.0/4.0 Aug 2024 - May 2026
- **Sharif University of Technology** Tehran, Iran
Bachelor of Computer Engineering; GPA: 19.03/20 Sep 2019 - Feb 2024

RESEARCH INTERESTS

- LLM Post-Training
- Multimodal LLMs
- LLM Efficiency
- Agentic AI and Tool Use

PUBLICATIONS

** denotes equal contribution.*

- "Learning to Stop without Learning to Stop: Self-Supervised Confidence Training Improves Reasoning Efficiency"
P Hosseini, A Tigalappanavara, S Nawathe, C Fan, S Basu, GI Winata, A Das, S Feizi, N Chitsazan *Preprint*
- "Early Stopping for Large Reasoning Models via Confidence Dynamics"
P Hosseini, S Nawathe*, M Salmani*, M Razaviyayn, S Feizi* *COLM 2026 Workshop (Spotlight)*
- "GHOST: Hallucination-Inducing Image Generation for Multimodal LLMs"
AY Parast, P Hosseini*, H Asadollahzadeh, AS Moakhar, B Azam, S Feizi, N Akhtar* *ICLR 2026*
- "SpurLens: Automatic Detection of Spurious Cues in Multimodal LLMs"
P Hosseini, S Nawathe*, M Moayeri, S Balasubramanian, S Feizi* *TMLR*
- "Your LLM Agents are Temporally Blind: The Misalignment Between Tool Use Decisions and Human Time Perception"
Y Cheng, AS Moakhar*, C Fan*, P Hosseini, K Faghieh, Z Sodagar, W Wang, S Feizi* *ACL 2026*
- "Gaming Tool Preferences in Agentic LLMs"
K Faghieh, W Wang*, Y Cheng*, S Bharti, G Sriramanan, S Balasubramanian, P Hosseini, S Feizi* *EMNLP 2025*
- "Chain-of-defensive-thought: Structured Reasoning Elicits Robustness in LLMs against Reference Corruption"
W Wang, P Hosseini*, S Feizi* *Preprint*
- "Decompose-and-Compose: A Compositional Approach to Mitigating Spurious Correlation"
FH Noohdani, P Hosseini, AY Parast*, HY Araghi, MS Baghshah* *CVPR 2024*

RESEARCH EXPERIENCE

- **Graduate Research Assistant, Reliable AI Lab, University of Maryland** College Park, Maryland
Advisor: Soheil Feizi Aug 2024 - Present
 - **Research Topics:** Efficient LLM Reasoning, Multimodal LLMs, Tool Use in Agentic AI
- **Intern, Capital One** Manhattan, New York
Mentor: Nima Chitsazan Jun 2026 - Sep 2026
 - **Research Topics:** Efficient Reasoning, Post-training LLM
- **Intern, Okinawa Institute of Science and Technology** Okinawa, Japan
Mentors: Makoto Yamada, Mohammad Sabokrou Jun 2023 - Sep 2023
 - **Research Topics:** Trustworthy ML, Continual Learning

HONORS, AWARDS, AND TALKS

- **Dean's Fellowship**, Department of Computer Science, University of Maryland Sep 2024, 2025
- **Talk:** "Mitigating Spurious Correlation", Okinawa Institute of Science and Technology Sep 2023
- **Gold medal in Iranian Mathematical Olympiad** Sep 2018
- **Silver medal in Iranian Geometry Olympiad** Sep 2018

SERVICE

- **ICLR Workshop Co-organizer:** Workshop on Spurious Correlation and Shortcut Learning (ICLR 2025)
- **Reviewer:** Neurips, ICLR, ICML, ACL, COLM

SKILLS SUMMARY

- **Programming:** Python, C/C++, UNIX Shell, Java, SQL, LaTeX
- **Machine Learning Frameworks:** PyTorch, TensorFlow, Hugging Face
- **AI/ML Expertise:** Multimodal LLMs, Diffusion Models, CLIP, Robustness, Spurious Correlation
- **Systems & Infrastructure:** MCP, GPU Clusters, SLURM, Virtual Machines, PostgreSQL, Git, Jupyter

LANGUAGES

- English: Fluent
- Persian: Native
- Japanese: Elementary Proficiency

TEACHING EXPERIENCE

- Teaching Assistant**

2021–2024

Teaching Assistant for major CS courses including Machine Learning, Artificial Intelligence, Convex Optimization, Compiler Design, Probability & Statistics, and Discrete Structures.
- Math Olympiad Instructor**

2019–2024

Taught problem-solving and competition math to high school students; several achieved top national ranks.

HOBBIES AND INTERESTS

- Calisthenics, Weight lifting